ISA/IEC 62443-3-3 / 4-2 • SL-3 TESTING CHEAT SHEET

Red Lion / N-Tron N-View (OPC Server & Diagnostic Monitor / N-View 2) · Component type: **Software Application (SAR) - network monitoring / management software** · Target SL = SL 3

Where to work	**N-View** on a Windows host: the **N-View OPC Server** + **Diagnostic Monitor** (per-port stats to OPC clients/HMIs); **N-View 2** adds SNMP auto-discovery + **centralized firmware management**. Switch configuration is done on the switch web UI (see the N-Tron 700 sheet) - N-View is monitoring + firmware push. Installer from Red Lion.
Ports / paths	N-View collects via the **N-View protocol** (Layer-2 multicast from N-View firmware) and **SNMP UDP 161** (N-View 2 discovery); serves data via **OPC DA (DCOM: RPC 135 + dynamic)** to OPC clients; firmware push via **TFTP/HTTP**. No TLS on the legacy OPC-DA/N-View path. Runs on Windows - harden the host.
Notes	SAR - Red Lion/N-Tron N-View monitoring software. It is largely **read-only monitoring**; the security-relevant write path is **N-View 2 firmware management** (it can push switch firmware - control who may and verify provenance). Legacy **OPC DA** (DCOM) + **SNMP** are the exposure. Confirm the N-View version; prefer SNMPv3 against the switches (configured on the switch).
How to read	Each row = native step-by-step (N-View / host) + command-level external-tool steps, noting where control is **delegated to the switch or host**. Default read-only. Mark Pass/Fail/Partial/N/A. Write/exploit tools = isolated env + written authorization.

FR 1 — Identification & Authentication Control (IAC)

Req (3-3 / 4-2)	Requirement	Where & how to verify (native steps + Ext. tools)	Pass criteria / evidence	Notes / version
SR/CR 1.1	Human user identification & auth	N-View has **limited app-level login**; authentication is enforced (a) at the **host OS** - only authorized Windows users run N-View / the OPC server - and (b) at the **switch** - SNMP credentials (prefer v3) and the switch web login gate configuration. Verify both; record the N-View-level residual. Restrict the OPC server so only authorized OPC clients connect (DCOM permissions). **Test:** confirm an unauthorized OPC client / Windows user cannot read the OPC server or push firmware. **Ext. tools:** Host (who may run N-View); DCOM config (OPC client permissions); switch SNMPv3 creds (per switch sheet).	Host + switch enforce access; OPC/DCOM permissions restrict clients; N-View residual documented.	Partial/Compensating.
SR/CR 1.2	Software process / device identification	Confirm each monitored switch is uniquely identified in N-View (IP/MAC/name from discovery) and N-View monitors only the intended devices; SNMP engine IDs (v3) identify the switches. **Ext. tools:** N-View device list; `nmap -sU -p 161 --script snmp-info <switch>`; Wireshark (N-View multicast / SNMP).	Unique switch identities in N-View; no rogue devices monitored.	
SR/CR 1.3	Account management	N-View has no user database - manage the host OS accounts that run it (least privilege), who may perform N-View 2 firmware pushes, and the switch SNMP/web accounts (per the switch sheet). No shared SNMP communities. **Ext. tools:** Host accounts; N-View firmware-push authorization; switch SNMP/web accounts.	Host + switch accounts managed/least-privilege; firmware-push restricted.	Compensating.
SR/CR 1.4	Identifier management	Confirm unique host user IDs and unique switch identities (IP/SNMP engine ID); no reuse. **Ext. tools:** Host/AD; N-View device list; inventory.	Unique host-user and switch identifiers; no reuse.	
SR/CR 1.5	Authenticator management	The authenticators are the switch SNMPv3 credentials + web password (strong, non-default - per the switch sheet) and the host/AD password policy. Avoid SNMP v1/v2 community strings; if legacy N-View requires v1/v2, treat the community as a secret + compensate with segmentation. **Ext. tools:** Switch SNMPv3 config; host/AD policy; confirm v1/v2 communities not default.	Strong SNMPv3 + host credentials; defaults changed; v1/v2 avoided/compensated.	Partial (legacy SNMP).
SR 1.6	Wireless access management	**N/A.** Monitoring software with no wireless interface. **Ext. tools:** Confirm no device wireless; host WLAN governed by OS/network.	N/A - justification recorded (software).	N/A
SR/CR 1.7	Strength of password-based authentication	Confirm the password-strength policy (length/complexity/lifetime) meets SL-3. **Steps:** for this product this is assessed with the same evidence as **SR/CR 1.5** above (shared steps); confirm this requirement individually and adjudicate it separately.	Strength of password-based authentication verified individually.	

Req (3-3 / 4-2)	Requirement	Where & how to verify (native steps + Ext. tools)	Pass criteria / evidence	Notes / version
SR/CR 1.8	Public key infrastructure (PKI) certificates	**N/A / limited.** Legacy OPC DA / N-View / SNMP have no TLS/PKI. Record N/A; provide confidentiality/integrity via segmentation and host controls; where SNMPv3 is used, its auth/priv (SHA/AES) is the crypto. **Ext. tools:** Confirm no TLS on OPC-DA/N-View; SNMPv3 auth/priv (SHA/AES) in use.	N/A/limited PKI; SNMPv3 crypto where available; compensating segmentation.	N/A/Partial.
SR/CR 1.9	Strength of public key-based authentication	Confirm public-key authentication uses strong, current keys/algorithms. **Steps:** for this product this is assessed with the same evidence as **SR/CR 1.8** above (shared steps); confirm this requirement individually and adjudicate it separately.	Strength of public key-based authentication verified individually.	
SR/CR 1.10	Authenticator feedback	Confirm N-View / SNMP credential entry masks secrets (no cleartext echo); note SNMP v1/v2 community strings traverse the wire in cleartext -> use v3. **Ext. tools:** Visual verification; Wireshark (confirm no cleartext v1/v2 communities on critical links).	Authenticators obscured; cleartext SNMP avoided (v3).	
SR/CR 1.11	Unsuccessful login attempts	Lockout is a switch/host concern, not N-View. The switch logs an **Authentication Error trap** on failed access - forward it; host/AD lockout for the N-View workstation. **Ext. tools:** Switch SNMP Authentication-Error trap to the SIEM; host/AD lockout.	Auth failures trapped/forwarded; host lockout; N-View residual noted.	Delegated.
SR/CR 1.12	System use notification	No N-View banner; enforce the approved use notice on the host (GPO). **Ext. tools:** Host `gpresult /h gp.html`.	Use notice at the host (compensating).	Compensating.
SR 1.13	Access via untrusted networks	Confirm the N-View host + switches are in the trusted OT/management zone; the OPC/DCOM + SNMP + N-View multicast are not exposed to untrusted networks; remote access via VPN/jump host. **Ext. tools:** `nmap -Pn -p 135,161 <host/switch>` from the untrusted side (expect filtered); firewall review.	N-View/OPC/SNMP unreachable from untrusted networks; remote via VPN.	

FR 2 — Use Control (UC)

Req (3-3 / 4-2)	Requirement	Where & how to verify (native steps + Ext. tools)	Pass criteria / evidence	Notes / version
SR/CR 2.1	Authorization enforcement	N-View is read-mostly; the write path is **N-View 2 firmware management** - confirm only authorized operators can push firmware (host account + N-View controls), and OPC clients get read-only data. The switch enforces config authorization (web/SNMP). **Test:** as an unauthorized user, attempt an N-View 2 firmware push -> denied. **Ext. tools:** N-View (firmware-push authorization); DCOM (OPC read-only); switch config rights.	Firmware push restricted; OPC read-only; switch config authorization enforced.	Partial/Delegated.
SR/CR 2.2	Wireless use control	**N/A.** Software, no wireless interface. **Ext. tools:** Confirm no wireless.	N/A - justification recorded.	N/A
SR/CR 2.3	Use control for portable/mobile devices	Control the N-View laptop and removable media (OS device-control); only authorized firmware images loaded for N-View 2 pushes; secure the host. **Ext. tools:** Host removable-media GPO; firmware-image provenance.	Portable/engineering media controlled; firmware images verified.	
SR/SAR 2.4	Mobile code	No general-purpose mobile code; confirm only signed N-View runs and no untrusted add-ons; validate the installer. **Ext. tools:** Installed-component review; verify installer signature.	Only signed/required components; no untrusted code.	SAR-specific.
SR/CR 2.5	Session lock	Confirm the host OS screen lock on inactivity; N-View re-auth via the host + switch SNMP creds. **Ext. tools:** Host inactivity-lock GPO.	Host locks on inactivity.	Compensating.
SR/CR 2.6	Remote session termination	Confirm OPC client connections can be dropped and are bounded/monitored (DCOM); N-View 2 firmware sessions are single/controlled. **Ext. tools:** OPC server session review; DCOM connection monitoring.	Sessions terminable; concurrency bounded/monitored.	
SR/CR 2.7	Concurrent session control	Confirm concurrent sessions are bounded/limited and monitored. **Steps:** for this product this is assessed with the same evidence as **SR/CR 2.6** above (shared steps); confirm this requirement individually and adjudicate it separately.	Concurrent session control verified individually.	

Req (3-3 / 4-2)	Requirement	Where & how to verify (native steps + Ext. tools)	Pass criteria / evidence	Notes / version
SR/CR 2.8	Auditable events	N-View has limited native audit; capture firmware-push actions on the host and forward switch **SNMP traps** (Authentication Error, Cold/Warm Start, Link Up/Down) to the SIEM; host audit for who ran N-View. **Ext. tools:** Host audit (firmware pushes); switch SNMP traps -> SIEM.	Firmware actions + switch traps captured/forwarded; host attribution.	Compensating.
SR/CR 2.9	Audit storage capacity	Logging is off-device (host + SIEM) - confirm sizing/retention and alerting on collection failure. **Ext. tools:** SIEM retention + source-silent alert.	Logs sized/offloaded; audit-failure alerting defined.	
SR/CR 2.10	Response to audit processing failures	Confirm a defined response to audit-processing/forwarding failures (alerting; no silent loss). **Steps:** for this product this is assessed with the same evidence as **SR/CR 2.9** above (shared steps); confirm this requirement individually and adjudicate it separately.	Response to audit processing failures verified individually.	
SR/CR 2.11	Timestamps	Confirm the host + switches are time-synced (NTP; switch SNTP) so trap/monitor timestamps correlate. **Ext. tools:** `w32tm /query /status`; switch time (per switch sheet).	Accurate, synchronized timestamps.	
SR/CR 2.12	Non-repudiation	Confirm actions are attributable to a specific user (non-repudiation). **Steps:** for this product this is assessed with the same evidence as **SR/CR 2.8** above (shared steps); confirm this requirement individually and adjudicate it separately.	Non-repudiation verified individually.	

FR 3 — System Integrity (SI)

Req (3-3 / 4-2)	Requirement	Where & how to verify (native steps + Ext. tools)	Pass criteria / evidence	Notes / version
SR/CR 3.1	Communication integrity	N-View OPC-DA / N-View multicast / SNMP v1/v2 are **unauthenticated cleartext** - contain them in a segmented management zone; use **SNMPv3** (auth+priv) where the switch supports it; do not expose OPC/DCOM across zone boundaries. **Ext. tools:** Wireshark (OPC/N-View/SNMP; confirm in-zone + v3 where used); firewall review.	Monitoring comms contained/segmented; SNMPv3 where supported; no crossing the boundary.	Legacy - Compensating.
SR/SAR 3.2	Protection from malicious code (component)	Confirm N-View software integrity: install only the Red Lion-**signed** N-View (verify the installer signature / download from Red Lion); host anti-malware; restrict install to admins. Verify switch firmware images (for N-View 2 pushes) are Red Lion-provenance. **Ext. tools:** `signtool verify /pa <setup>.exe`; host AV; firmware-image provenance.	Signed software validated; install restricted; firmware images verified.	SAR-specific.
SR/CR 3.3	Security functionality verification	After any change re-verify: DCOM/OPC client permissions, SNMPv3 use, firmware-push authorization, and patch level, against the hardening baseline; confirm the monitored switch security still matches its baseline (per the switch sheet). **Ext. tools:** N-View/DCOM config review; switch config compare; version check.	Security-relevant settings match the baseline after change.	
SR/CR 3.4	Software & information integrity	Protect the N-View configuration/site data and any stored SNMP credentials/firmware images on the host: ACLs, encrypt at rest (BitLocker), back up under change control. **Ext. tools:** Host ACL on N-View data; confirm no plaintext SNMP creds; backup review.	N-View data protected; credentials/images secured.	
SR/CR 3.5	Input validation	Patch known N-View / OPC CVEs; confirm N-View handles malformed switch responses without crashing. **ISOLATED:** test against a bench switch. **Ext. tools:** Nessus/OpenVAS on the host; Red Lion/CISA advisory cross-check; ISOLATED malformed-response test.	Known CVEs patched; malformed input handled safely.	
SR/CR 3.6	Deterministic Output	**N/A.** Device does not drive process outputs; deterministic/fail-safe output is not applicable.	N/A - justification recorded (not assessed as a control).	N/A
SR/CR 3.7	Error handling	Confirm N-View handles errors gracefully without leaking sensitive info (no plaintext SNMP community/credential in logs). **Ext. tools:** N-View / host log review.	Graceful error handling; no credential leak.	
SR/CR 3.8	Session integrity	**N/A / limited.** Legacy OPC-DA/N-View/SNMPv1-2 sessions are not secured; SNMPv3 sessions are integrity-protected. Contain the rest by segmentation. **Ext. tools:** Wireshark (SNMPv3 vs cleartext).	SNMPv3 sessions integrity-protected; legacy contained.	N/A/Partial.

Req (3-3 / 4-2)	Requirement	Where & how to verify (native steps + Ext. tools)	Pass criteria / evidence	Notes / version
SR/CR 3.9	Protection of audit information	Protect the host/SIEM audit stores; restrict who can read/clear them (host ACLs). **Ext. tools:** Host ACL; SIEM access review.	Audit records protected from unauthorized read/modify/delete.	

FR 4 — Data Confidentiality (DC)

Req (3-3 / 4-2)	Requirement	Where & how to verify (native steps + Ext. tools)	Pass criteria / evidence	Notes / version
SR/CR 4.1	Information confidentiality	Confirm information confidentiality — sensitive data is protected in transit and at rest. **Steps:** for this product this is assessed with the same evidence as **SR/CR 3.1** above (shared steps); confirm this requirement individually and adjudicate it separately.	Information confidentiality verified individually.	
SR/CR 4.2	Information persistence	On decommission/repurpose of the host, remove the N-View config/site data, stored SNMP credentials, firmware images, and backups; confirm no plaintext secrets persist; encrypt the host disk. **Ext. tools:** grep for secrets; confirm data removed; BitLocker status.	N-View data/credentials purged on decommission; disk encrypted.	
SR/CR 4.3	Use of cryptography	N-View has no native transport crypto (OPC DA/N-View); use SNMPv3 (SHA/AES) against the switches and rely on host disk encryption + segmentation. Protect stored credentials. **Ext. tools:** Confirm SNMPv3 auth/priv; host encryption; no plaintext creds.	SNMPv3 crypto where available; host encryption; compensating segmentation.	Partial.

FR 5 — Restricted Data Flow (RDF)

Req (3-3 / 4-2)	Requirement	Where & how to verify (native steps + Ext. tools)	Pass criteria / evidence	Notes / version
SR/CR 5.1	Network segmentation	Confirm the N-View host + switches sit in a segmented management/OT zone behind a firewall; OPC/DCOM + SNMP + N-View multicast do not cross to IT; remote use via a jump host. **Ext. tools:** `nmap -p 135,161 <host/switch>` per zone; firewall rule review.	Host + switches in a segmented zone; only required flows cross.	
SR 5.2	Zone boundary protection	Confirm zone-boundary protection — only required flows cross the boundary (system SR / network device). **Steps:** for this product this is assessed with the same evidence as **SR/CR 5.1** above (shared steps); confirm this requirement individually and adjudicate it separately.	Zone boundary protection verified individually.	
SR 5.3	General-purpose person-to-person comms restrictions	**N/A.** Monitoring software with no general-purpose person-to-person messaging. **Ext. tools:** Confirm no such services.	N/A - justification recorded.	N/A
SR 5.4	Application partitioning	Partition by criticality: separate N-View instances/zones and least-privilege host access; do not monitor unrelated criticality zones from one flat instance. **Ext. tools:** N-View instance/zone segregation; host access review.	Monitoring scopes partitioned by criticality.	

FR 6 — Timely Response to Events (TRE)

Req (3-3 / 4-2)	Requirement	Where & how to verify (native steps + Ext. tools)	Pass criteria / evidence	Notes / version
SR/CR 6.1	Audit log accessibility	This tool IS a monitoring layer: confirm N-View traffic/status + switch SNMP traps are readable/exportable and forwarded to the SIEM; N-View dashboards (or Crimson) provide continuous monitoring; passive OT IDS on the segment. **Ext. tools:** N-View dashboard/export; switch traps -> SIEM; passive OT IDS.	Events accessible/exportable and monitored in near-real-time.	

Req (3-3 / 4-2)	Requirement	Where & how to verify (native steps + Ext. tools)	Pass criteria / evidence	Notes / version
SR/CR 6.2	Continuous monitoring	Confirm continuous monitoring — events forwarded to and monitored by a SIEM/OT-IDS in near-real-time. **Steps:** for this product this is assessed with the same evidence as **SR/CR 6.1** above (shared steps); confirm this requirement individually and adjudicate it separately.	Continuous monitoring verified individually.	

FR 7 — Resource Availability (RA)

Req (3-3 / 4-2)	Requirement	Where & how to verify (native steps + Ext. tools)	Pass criteria / evidence	Notes / version
SR/CR 7.1	Denial of service protection	Confirm the N-View host is resourced/hardened and the OPC/SNMP polling load is bounded so a flood cannot deny monitoring; do not co-host unrelated apps; the N-View multicast load is contained to the management VLAN. **Ext. tools:** Host resource monitor; polling-load review; ISOLATED load test.	Resources bounded; monitoring available under load.	
SR/CR 7.2	Resource management	Confirm resource management — resources are bounded; no exhaustion-based loss of function. **Steps:** for this product this is assessed with the same evidence as **SR/CR 7.1** above (shared steps); confirm this requirement individually and adjudicate it separately.	Resource management verified individually.	
SR/CR 7.3	Control system backup	Confirm the N-View config/site data + switch firmware images are backed up (versioned, offline) and a restore/rebuild is tested. **Ext. tools:** Backup repository; restore test on a lab host.	Backups current and recovery validated.	
SR/CR 7.4	Control system recovery & reconstitution	Confirm recovery/reconstitution — a restore to a known-good state is tested. **Steps:** for this product this is assessed with the same evidence as **SR/CR 7.3** above (shared steps); confirm this requirement individually and adjudicate it separately.	Control system recovery & reconstitution verified individually.	
SR/CR 7.5	Emergency Power	**N/A.** Emergency/backup power is provided at the facility/UPS level, not the device; confirm the device retains its security state across a power transfer.	N/A - justification recorded (not assessed as a control).	N/A
SR/CR 7.6	Network & security configuration settings	Harden: run N-View on a hardened least-privilege host, install only N-View, keep it patched, restrict DCOM/OPC to authorized clients, prefer SNMPv3 (disable v1/v2 on the switches), and remove sample data. **Ext. tools:** Host/DCOM hardening review; N-View version; switch SNMP config.	Least functionality; DCOM/SNMP restricted; patched.	
SR/CR 7.7	Least functionality	Confirm least functionality — unused services/ports/protocols are disabled. **Steps:** for this product this is assessed with the same evidence as **SR/CR 7.6** above (shared steps); confirm this requirement individually and adjudicate it separately.	Least functionality verified individually.	
SR/CR 7.8	Component inventory	Inventory the N-View version, the host OS, and the monitored switches (model/firmware) so vulnerabilities track against Red Lion / CISA advisories. **Ext. tools:** N-View version; host inventory; Red Lion/CISA feeds.	Accurate version + monitored-switch inventory maintained; advisories tracked.	

External Test Tools — how to use

Tool / method	Verifies	How to use — steps / example	Constraints
N-View (OPC Server + Diagnostic Monitor / N-View 2)	Native monitoring & firmware review (all FRs)	Review the OPC server config (client/DCOM permissions), the discovered switch list, dashboards, and (N-View 2) firmware-management authorization. Switch config is on the switch sheet.	Read/authorized; firmware push authorized only.
Host OS controls	Who runs N-View + protect data (1.1, 2.1, 3.4, 4.2)	App-execution control, DCOM permissions (OPC clients), removable-media policy, ACLs + BitLocker on the N-View data.	Windows admin review.
signtool	Signed-installer validation (SAR 3.2)	`signtool verify /pa <setup>.exe`; download from Red Lion.	Read-only.

Tool / method	Verifies	How to use — steps / example	Constraints
nmap	Switch/service exposure & segmentation (1.13, 5.x)	`nmap -sU -p 161 --script snmp-info <switch>`; `nmap -p 135 <host>`; scan per zone.	Read-only; authorized.
Wireshark	Comms integrity & SNMP (1.10, 3.1, 3.8)	Confirm N-View multicast/OPC/SNMP stay in-zone; SNMPv3 (not cleartext v1/v2) on management.	Passive; SPAN/tap only.
Nessus / OpenVAS + Red Lion/CISA advisories	CVE scan (3.5, 7.8)	Scan the N-View host for known CVEs; cross-check Red Lion / CISA advisories.	ISOLATED/representative; authorization.

Controlled - Project Use · Independent Assurance Services · ISA/IEC 62443-3-3 / 4-2 · Product-specific companion to the 62443 assessment tool; confirm exact menu paths / CLI against the deployed firmware and the project hardening baseline. ISOLATED tools run only in a representative/lab environment with written authorization.